Gaurav Singh
Structural & Computational Engineer
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Education

Degree	**Institution**	**CPI/%**	**Year**
Masters, Structural	IIT Gandhinagar, Gujarat	7.5	2025
Bachelors, Civil	IIT Palakkad, Kerala	7	2021

Work Experience

R&D Engineer
Modulus housing Pvt. Ltd. | *Sept 2025–Present*

- Led research and development in finite element simulations, cost optimization of reinforced concrete structural elements (beams, columns, footings and wall panels) using Abaqus and custom codes, achieving significant material and cost efficiency improvements compared to traditional prismatic elements.
- Developed a **MATLAB-based nonlinear beam optimization framework** incorporating provisions from IS 456, using a segment-wise optimization strategy based on bending moment and shear force distributions. Achieved **40–50% cost reduction** while satisfying strength, serviceability, and safety constraints.
- Designed **parametric and manufacturable CAD models** of optimized structural elements and their casting moulds using SolidWorks, directly enabling fabrication and experimental validation.
- Performed advanced **nonlinear finite element simulations** using the **Concrete Damage Plasticity (CDP)** model in Abaqus, incorporating material, geometric and contact non linearities in modelling.
- Conducted **FEM simulations** of modular reinforced concrete bathroom pods included Reinforced concrete with embedded steel rebars, Hyperelastic materials (neoprene rubber supports), Hyperfoam EPS components, Nonlinear contact behavior, considered Dynamic loading scenarios (pod crane lifting,pod truck transportation, vibrations, modal analysis, impact loading etc) Ensured complete structural safety during handling and transport.
- Simulated and validated **TRC (Textile Reinforced Concrete) panels and EPS wall systems**, validated beams, columns simulation via research papers achieving strong correlation between numerical and experimental results.
- Performed **global structural analysis** by converting AutoCAD drawings into STAAD.Pro models; extracted force envelopes and stress distributions to drive optimization workflows.
- Analyzed a wide range of structural systems which can potentially reduce cost of construction, including:
  - Topology-optimized beams
  - Variable cross-section (tapered) beams
  - Shell-based optimized structural elements
  - Load-path optimized beams
  - Columns, and footings
  - Whole 3D pods
- Did validation of FEM models through IS456 standards through PM interaction (columns) and moment curvature (beams) achieved good correlation with abaqus solver results hand design calculations, developed strong expertise in nonlinear FEM.
- Worked with high fidelity simulations of UPHC and FRC developed material properties generator using fiber model code and euro code standards and research papers approximated equations, validated developed material model with experimental papers and explicit fiber modelling in Abaqus.

CAX Software Engineer (Technical Specialist)
Acquisy Solutions Pvt. Ltd. | *July 2024–Sept 2025*

Working on Altair's AEC solutions: *s-frame, s-concrete* and *s-foundation* ad-ins software development.

- Developed and validated a wide range of **structural design calculation frameworks** using Maple Flow for beams, columns, slabs, and walls; cross-verified results against textbooks, Excel tools, and commercial software such as ETABS and STAAD.Pro.
- Implemented reinforced concrete design provisions from IS 456 and IS 13920 into **s-concrete** using **VB.NET and C#**, ensuring accuracy through rigorous validation with Maple Flow calculation canvases.
- Led development of **IS 456 and KDS 14 20 (Korean Design Code)** modules in **s-concrete**, enabling adoption of the software by structural engineers across India and South Korea.
- Performed detailed comparative studies of international design standards, including ACI 318 and Eurocode 2, with a focus on **P–M interaction curves**, stress block parameters, and design philosophies across Indian and Korean codes.
- Contributed to feature development in **s-concrete**, including:
  - Design report generation and technical summaries
  - UI/UX enhancements using **C# WPF**
  - Development of technical help documentation for end users
- Played a key role in **S-Foundation module development**, validating both **rigid and flexible finite element formulations** for:
  - Isolated, eccentric, combined, raft, wall, and pile foundations
  - Soil–structure interaction on elastic foundations. Validation was performed using **hand calculations and benchmark problems**.
- Enhanced interoperability of **s-frame** with external structural software such as STAAD.Pro, ETABS, and MIDAS by:
  - Reverse engineering proprietary file formats
  - Developing robust import/export pipelines using **C# and WPF**
- Collaborated with international teams including Altair Engineering (Canada) through regular technical discussions to gather requirements and propose engineering solutions; worked closely with Korean engineers and academic experts to align software features with regional codes and practices.

- Developed **custom Python scripts** for advanced mesh and simulation data processing, including automated extraction and post-processing of nodal forces and results tailored to client-specific requirements.

Project Associate (Part time alongside college)
Indian Institute of Technology (IIT) Gandhinagar | *Jan 2023–July 2024*
Worked on a consultancy project for egress modeling of Bangalore metro stations

- Developed advanced **fire dynamics and pedestrian evacuation simulations** using FDS+Evac, integrating detailed CAD layouts of metro stations to evaluate **emergency evacuation performance under fire scenarios**.
- Conducted an extensive literature review and generated **150+ simulation scenarios**, automating case creation and execution through **custom MATLAB scripting**, enabling large-scale parametric analysis of evacuation conditions in metro infrastructure.
- Applied **Design of Experiments (DOE)**, statistical analysis, and **Explainable Machine Learning (XAI)** techniques to quantify the influence of **15+ fire and occupant-related parameters** (e.g., fire growth rate, visibility, crowd density, behavioral response) on evacuation outcomes.
- Identified **optimal parameter combinations** representing realistic Indian crowd behavior, improving the fidelity of evacuation simulations for metro stations.
- Leveraged machine learning insights to **calibrate occupant movement models**, enhancing prediction accuracy under Indian demographic and behavioral conditions.
- Evaluated key safety metrics including:
    - Evacuation time and bottleneck formation
    - Smoke spread and visibility effects
    - Occupant flow rates and congestion patterns
- Published this work as a **master's thesis** and presented findings at the International Conference on Computational Mechanics and Simulations (ICCMS).

Teaching Assistant
Mechanical Workshop, IIT Gandhinagar | *Aug 2022–Dec 2022*

- Taught **first-year undergraduate students** the operation of mechanical workshop machinery, including **electric arc welding** and **cutting/fitting techniques**.
- Ensure students' safety while guiding hands-on training.

Personal/academic Projects and other technical works

Topology Optimization of 2D Plates with Large Deformations (Personal Project)

- Developed a **nonlinear FEM-based topology optimization code** in MATLAB for Mindlin plates using the SIMP method.
- Implemented **5 DOFs per node (u, v, w, θx, θy)** with bilinear shape functions.
- Solved geometric nonlinearity using the **Newton–Raphson method**.
- Investigated the effect of large deformations on optimal material distribution.

Design and Analysis of G+5 Structure (Academic Project)

- Performed structural analysis using SAP2000 and MATLAB.
- Calculated wind loads (IS 875) and seismic loads (IS 1893).
- Designed beams, columns, slabs, shear walls, and footings as per IS 456 and IS 13920.

Experimental & Numerical Study of 3D Truss Structures (Academic Project)

- Fabricated and tested a 3D truss under loading using CTM to obtain load–displacement response.
- Developed MATLAB code using the **stiffness matrix method with sub structuring and static condensation**.
- Validated results against Open Sees.

Dynamic Analysis of MDOF Structures (Academic Project)

- Designed a **2-DOF experimental structure** with Arduino-controlled excitation system.
- Measured acceleration, velocity, and displacement; applied signal filtering.
- Developed numerical model using Newmark-β method and validated with OpenSees

Two-Bar Truss Optimization (Academic Project)

- Formulated and solved a constrained optimization problem for minimum weight design.
- Considered yielding, buckling, and deflection constraints.
- Implemented solution using GEKKO in Python.

Failure Mode Prediction of Steel Tension Members (Bachelor's Thesis)

- Applied supervised machine learning to classify failure modes:
    - Gross section yielding
    - Net section fracture
    - Block shear failure
- Built dataset from experimental studies and predicted **failure mode probabilities**.

Fire Evacuation Parameter Optimization (master's Research)

GUI Applications using Tkinter (Personal Project)

- Developed Python-based GUI tools:
  - Scientific calculator with multiple modes
  - 2-octave virtual piano
- Published project on GitHub.

Construction Planning of Emergency Hospital (Academic Project)

- Planned and optimized construction schedule using Primavera.
- Modeled after rapid construction of emergency hospitals during COVID-19.
- Achieved optimized project completion timeline of **47 days**.

Skill Summary

Languages: Worked with Python, MATLAB, C++, C#, VB.NET
Tools used: FDS+EVAC, SAAP2000, Sagemath, Revit, Etabs, opensees, Autocad, ANSYS, Primavera, Minitab, Autocad, pyrosim, pathfinder, s-frame, s-foundation, s-concrete staad pro, solidworks, Abaqus, Staad pro
Relevant courses: FEM, CFD, Numerical methods, Structural analysis, dynamics and solid mechanics, machine learning and Optimization methods for engineering, Structural design for fire.

Research papers publications/conferences

- **"Numerical Investigations on Evacuation Performance of Passengers in Underground Metro Stations"** Presented at the **9th International Congress on Computational Mechanics and Simulation** (December 20–22, 2023, IIT Gandhinagar, India).
- Presented work to metro industry professionals from metro stations across India on metro safety work at a small workshop in IIT Gandhinagar (Jan 2024).

Positions of responsibilities/volunteering activities

Evidyaloka Teaching Volunteer:

- Taught mathematics to underprivileged rural children in India via Skype.
- Contributed to **content and video development** for educational material during the COVID-19 pandemic.

IITGN Neev Volunteer:

- Conducted sessions on **MS Office and computer hardware** for graduates in Basan village and supervised their training.
- **Chegg Subject Matter Expert**: Solved and explained civil engineering problems for students worldwide.

Unnat Bharat Abhiyan:

- Collaborated with a team of four to **survey rural residential complexes** in Kerala, gathering relevant data for development projects.

Hobbies and Interests

- **Hobbies:** Playing football, sketching, and playing the guitar.
- **Interests:** Passionate about structural engineering, mathematics, and physics. Often spends free time learning about these fields.